BIANCA

175 mm

BLOPRESS PLUS TABLET 16/12.5mg

(Candesartan cilexetil and Hydrochlorothiazide)

BLOPRESS PLUS is an oral combination product consisting of candesartan cilexetil, a selective angiotensin II (AII) subtype 1 (AT1) receptor antagonist, and hydrochlorothiazide (HCTZ), a thiazide diuretic. BLOPRESS PLUS has been developed as an antihypertensive.

COMPOSITION

BLOPRESS 16mg PLUS is a light pink, oval flat tablet with a score and engraved 16lC on both sides. Each tablet contains 16 mg candesartan cilexetil and 12.5 mg hydrochlorothiazide.

INDICATION

Treatment of essential hypertension where monotherapy with candesartan cilexetil or hydrochlorothiazide is not sufficient.

DOSAGE AND ADMINISTRATION

Dosage

The recommended dose of BLOPRESS PLUS is one tablet (16mg/12.5mg) once daily. When clinically appropriate a direct change from monotherapy to BLOPRESS PLUS may be considered.

Most of the antihypertensive effect is usually attained within 4 weeks of initiation of treatment.

Administration

BLOPRESS PLUS should be taken once daily with or without food.

Use in the elderly

No initial dosage adjustment is necessary in elderly patients.

Use in patients with intravascular volume depletion

Dose titration of candesartan cilexetil is recommended in patients at risk for hypotension, such as patients with possible volume depletion (an initial dose of candesartan cilexetil of 4 mg may be considered in these patients).

Use in impaired renal function

Loop diuretics are preferred to thiazides in this population. Dose titration of candesartan cilexetil is recommended in patients with renal impairment whose creatinine clearance is > 30 ml/min/1.73 m^2 BSA before treatment with BLOPRESS PLUS (the recommended starting dose of candesartan cilexetil is 4 mg in patients with mild to moderate renal impairment).

BLOPRESS PLUS should not be used in patients with severe renal impairment (creatinine clearance < 30 ml/min/1.73 m^2 BSA) (see CONTRAINDICATIONS).

Use in impaired hepatic function

Dose titration of candesartan cilexetil is recommended in patients with mild to moderate hepatic impairment before treatment with BLOPRESS PLUS (the recommended starting dose of candesartan cilexetil is 4 mg in these patients).

BLOPRESS PLUS should not be used in patients with severe hepatic impairment and/or cholestasis (see CONTRAINDICATIONS).

Use in children

The safety and efficacy of BLOPRESS PLUS have not been established in children.

CONTRAINDICATIONS

Hypersensitivity to any component of BLOPRESS PLUS or to sulfonamide derived drugs. Hydrochlorothiazide is a sulfonamide derived drug.

Pregnancy and lactation (see PREGNANCY AND LACTATION).

The use of BLOPRESS PLUS in combination with aliskiren-containing medicines in patients with diabetes or moderate to severe renal impairment (GFR<60 mL/min/1.73m^2) (see Interaction with Other Medications and Other Forms of Interaction).

Severe renal impairment (creatinine clearance < 30 ml/min/1.73 m^2 BSA) or anuria (see Special Warnings and Precautions).

Severe hepatic impairment and/or cholestasis.

Refractory hypokalaemia and hypercalcaemia.

Gout.

SPECIAL WARNINGS AND PRECAUTIONS

Dual blockade of the renin-angiotensin-aldosterone system (RAAS) with aliskiren-containing medicines

Dual blockade of the renin-angiotensin-aldosterone system (RAAS) by combining candesartan cilexetil/hydrochlorothiazide and aliskiren is not recommended since there is an increased risk of hypotension, hyperkalemia and changes in renal function (see Interaction with Other Medications and Other Forms of Interaction).

Renal artery stenosis

Renal function may worsen in patients with renal artery stenosis. Other drugs that affect the renin-angiotensin-aldosterone system, i.e. angiotensin converting enzyme (ACE) inhibitors, may increase blood urea and serum creatinine in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney. While this is not reported with BLOPRESS PLUS, a similar effect may be anticipated with angiotensin II receptor antagonists.

Intravascular volume depletion

In patients with intravascular volume and/or sodium depletion, symptomatic hypotension may occur, as described for other agents acting on the renin-angiotensin-aldosterone system.

Therefore, the use of BLOPRESS PLUS is not recommended until the hypovolemia has been corrected.

Renal impairment

Loop diuretics are preferred to thiazides in this population. When BLOPRESS PLUS is used in patients with impaired renal function, a periodic monitoring of potassium, creatinine and uric acid levels is recommended.

Kidney Transplantation

There is no experience regarding the administration of BLOPRESS PLUS in patients with recent kidney transplantation.

Anesthesia and surgery

Hypotension may occur during anesthesia and surgery in patients treated with angiotensin II antagonists due to blockade of the rennin-angiotensin system. Very rarely, hypotension may be severe such that it may warrant the use of intravenous fluids and/or vasopressors.

Hyperkalemia

Based on experience with the use of other drugs that affect the renin-angiotensin-aldosterone system, concomitant use of BLOPRESS PLUS and potassium-sparing diuretics, potassium supplements or salt substitutes or other drugs that may increase serum potassium levels (e.g. heparin sodium) may lead to increases in serum potassium in hypertensive patients.

Although not documented with BLOPRESS PLUS treatment with angiotensin converting enzyme inhibitors or angiotensin-II-receptor inhibitors may cause hyperkalemia, especially in the presence of heart failure and/or renal impairment.

Hepatic impairment

Thiazides should be used with caution in patients with impaired hepatic function or progressive liver disease, since minor alteration of fluid and electrolyte balance may precipitate hepatic coma. There is no clinical experience with BLOPRESS PLUS in patients with hepatic impairment.

Non-Melanoma Skin Cancer

An increased risk of non-melanoma skin cancer (NMSC) [basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)] with increasing cumulative dose of hydrochlorothiazide exposure has been observed in two epidemiological studies based on the Danish National Cancer Registry. Photosensitizing actions of hydrochlorothiazide could act as a possible mechanism for NMSC.

Patients taking candesartan cilexetil/hydrochlorothiazide should be informed of the risk of NMSC and advised to regularly check their skin for any new lesions and promptly report any suspicious skin lesions. Possible preventive measures such as limited exposure to sunlight and UV rays and, in case of exposure, adequate protection should be advised to the patients in order to minimize the risk of skin cancer. Suspicious skin lesions should be promptly examined potentially including histological examinations of biopsies. The use of candesartan cilexetil/hydrochlorothiazide may also need to be reconsidered in patients who have experienced previous NMSC.

Aortic and mitral valve stenosis and obstructive hypertrophic cardiomyopathy

As with other vasodilators, special caution is indicated in patients suffering from hemodynamically relevant aortic or mitral value stenosis, or obstructive hypertrophic cardiomyopathy.

Primary hyperaldosteronism

Patients with primary aldosteronism generally will not respond to antihypertensive drugs acting through inhibition of the renin-angiotensin system. Therefore, the use of BLOPRESS PLUS is not recommended in this population.

Electrolyte imbalance

Thiazides, including hydrochlorothiazide, can cause fluid or electrolyte imbalance (hypercalcemia, hypokalemia, hyponatremia, hypomagnesemia and hypochloremic alkalosis).

As for any patient receiving diuretic therapy, periodic determination of serum electrolytes should be performed at appropriate intervals.

Thiazide diuretics may decrease the urinary calcium excretion and may cause intermittent and slightly increased serum calcium concentrations. Marked hypercalcemia may be a sign of hidden hyperparathyroidism. Thiazides should be discontinued before carrying out tests for parathyroid function.

Hydrochlorothiazide dose-dependently increases urinary potassium excretion which may result in hypokalaemia. This effect of hydrochlorothiazide seems to be less evident when combined with candesartan cilexetil. The risk for hypokalaemia may be increased in patients with cirrhosis of the liver, in patients experiencing brisk diuresis, in patients with an inadequate oral intake of electrolytes and in patients receiving concomitant therapy with corticosteroids or adrenocorticotropic hormone (ACTH).

Thiazides have been shown to increase the urinary excretion of magnesium, which may result in hypomagnesaemia.

Metabolic and endocrine effects

Treatment with thiazide diuretics may impair glucose tolerance. Dosage adjustment of antidiabetic drugs, including insulin, may be required. Latent diabetes mellitus may become manifest during thiazide therapy. Increases in cholesterol and triglyceride levels have been associated with thiazide diuretic therapy; however at the 12.5 mg dose contained in BLOPRESS PLUS minimal or no effect was reported. Thiazide diuretics increase serum uric acid concentration and may precipitate gout in susceptible patients.

Hypersensitivity reactions

Hypersensitivity reactions to hydrochlorothiazide may occur in patients with or without a history of allergy or bronchial asthma, but are more likely in patients with such a history.

Systemic lupus erythematosus

Exacerbation or activation of systemic lupus erythematosus has been reported with the use of thiazide diuretics.

General

In patients whose vascular tone and renal function depend predominantly on the activity of the renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with other drugs that affect this system has been associated with acute hypotension, azotemia, oliguria or rarely, acute renal failure. The possibility of similar effects cannot be excluded with angiotensin II receptor antagonists. As with any antihypertensive agent, excessive blood pressure decrease in patients with ischaemic heart disease or atherosclerotic cerebrovascular disease could result in a myocardial infarction or stroke.

Intestinal angioedema

Intestinal angioedema has been reported in patients treated with angiotensin II receptor antagonists, candesartan cilexetil. (see section ADVERSE EFFECTS). These patients presented with abdominal pain, nausea, vomiting and diarrhoea. Symptoms resolved after discontinuation of angiotensin II receptor antagonists. If intestinal angioedema is diagnosed, candesartan cilexetil should be discontinued and appropriate monitoring should be initiated until complete resolution of symptoms has occurred.

SYMPTOMS AND TREATMENT OF OVERDOSE

Symptoms

Based on pharmacological considerations, the main manifestation of an overdose of candesartan cilexetil is likely to be symptomatic hypotension and dizziness. In single case reports of overdose up to 672 mg candesartan cilexetil in an adult, the patient recovery of patients was uneventful.

The main manifestation of an overdose of hydrochlorothiazide is acute loss of fluid and electrolytes. Symptoms such as dizziness, hypotension, thirst, tachycardia, ventricular arrhythmias, sedation/ impairment of consciousness and muscle cramps can also be observed.

Management

No specific information is available on the treatment of overdosage with candesartan cilexetil/ hydrochlorothiazide. The following measures are, however, suggested in case of overdosage. When indicated, induction of vomiting or gastric lavage should be considered. If symptomatic hypotension should occur, symptomatic treatment should be instituted and vital signs monitored. The patient should be placed supine with the legs elevated. If this is not sufficient, plasma volume should be increased by infusion of isotonic saline solution. Serum electrolyte and acid balance should be checked and corrected, if needed. Sympathomimetic drugs may be administered if the above-mentioned measures are not sufficient.

Candesartan cannot be removed by haemodialysis. It is not known to what extent hydrochlorothiazide is removed by haemodialysis.

INTERACTION WITH OTHER MEDICATIONS AND OTHER FORMS OF INTERACTION

Dual Blockade of the Renin-Angiotensin-Aldosterone System (RAAS)

Dual blockade of the RAAS with angiotensin receptor blockers, ACE inhibitors, or aliskiren is associated with increased risks of hypotension, hyperkalemia, and changes in renal function (including acute renal failure) compared to the use of a single RAAS-acting agent.

Lithium

Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with ACE inhibitors or hydrochlorothiazide. A similar effect may occur with angiotensin II receptor antagonists and careful monitoring of serum lithium levels is recommended during concomitant use.

Non-steroidal anti-inflammatory Drugs (NSAIDs)

Attenuation of the antihypertensive effect may occur when simultaneously administering angiotensin II receptor antagonists and NSAIDs; (i.e. selective COX-2 inhibitors, acetylsalicylic acid and non-selective NSAIDs). As with ACE inhibitors, concomitant use of angiotensin II receptor antagonists and NSAIDs may lead to an increased risk of worsening of renal function, including possible acute renal failure, and an increase in serum potassium, especially in patients with poor pre-existing renal function. The combination should be administered with caution, especially in older and volume-depleted patients. Patients should be adequately hydrated and consideration should be given to monitoring renal function after initiation of concomitant therapy and periodically thereafter.

The diuretic, natriuretic and antihypertensive effect of hydrochlorothiazide is blunted by NSAIDs.

Antihypertensives

The antihypertensive effect of candesartan cilexetil/hydrochlorothiazide may be enhanced by other medicinal products with blood pressure lowering properties.

Potassium-sparing diuretics, potassium supplements or salt substitutes containing potassium

Based on experience with the use of other medicinal products that affect the RAAS, concomitant use of candesartan cilexetil/hydrochlorothiazide and potassium-sparing diuretics, potassium supplements or salt substitutes or other drugs that may increase serum potassium levels (e.g. heparin sodium) may lead to increases in serum potassium.

Potassium-depleting drugs

The potassium depleting effect of hydrochlorothiazide could be expected to be potentiated by other drugs associated with potassium loss and hypokalemia (e.g. other kaliuretic diuretics, laxatives, amphotericin, carbenoxolone, penicillin G sodium, salicylic acid derivates).

Diuretic-induced hypokalemia and hypomagnesemia predisposes to the potential cardiotoxic effects of digitalis glycosides and antiarrhythmics. Periodic monitoring of serum potassium is recommended when candesartan cilexetil/hydrochlorothiazide is administered with such drugs.

Colestipol or cholestyramine

The absorption of hydrochlorothiazide is reduced by colestipol or cholestyramine.

Nondepolarizing skeletal muscle relaxants

The effect of nondepolarizing skeletal muscle relaxants (e.g. tubocurarine) may be potentiated by hydrochlorothiazide.

Adrenocorticotropic hormone

The risk for hypokalaemia may be increased during concomitant use of hydrochlorothiazide with steroids or ACTH.

Alcohol, barbiturates or anesthetics

Postural hypotension may become aggravated by simultaneous intake of hydrochlorothiazide with alcohol, barbiturates or anesthetics.

Antidiabetic drugs

Treatment with hydrochlorothiazide may impair glucose tolerance. Dosage adjustment of antidiabetic drugs, including insulin, may be required.

Beta-blockers and diazoxide

The hyperglycaemic effect of beta-blockers and diazoxide may be enhanced by thiazides.

Pressor amines

Hydrochlorothiazide may cause the arterial response of pressor amines (e.g. adrenaline) to decrease but not enough to exclude a pressor effect.

Calcium supplements or Vitamin D

Thiazide diuretics may increase serum calcium levels due to decreased excretion. If calcium supplements or Vitamin D must be prescribed, serum calcium levels should be monitored and dosage adjusted accordingly.

Anticholinergic agents

Anticholinergic agents (e.g. atropine, biperiden) may increase the bioavailability of thiazide-type diuretics by decreasing gastrointestinal motility and stomach emptying rate.

Amantadine

Thiazide may increase the risk of adverse effects caused by amantadine.

Cytotoxic drugs

Thiazides may reduce the renal excretion of cytotoxic drugs (e.g. cyclophosphamide, methotrexate) and potentiate their myelosuppressive effects.

Iodinated contrast media

Hydrochlorothiazide may increase the risk of acute renal insufficiency especially with high doses of iodinated contrast media.

PREGNANCY AND LACTATION

Use in Pregnancy

BLOPRESS PLUS must not be used in pregnancy (see CONTRAINDICATIONS). If pregnancy is detected during treatment, BLOPRESS PLUS must be immediately discontinued. Medicinal products that act directly on the renin-angiotensin-aldosterone system can cause fetal and neonatal injury (hypotension, renal dysfunction, oliguria and/or anuria, oligohydramnios, skull

VOLTA

hypoplasia, intrauterine growth retardation) and death.
Cases of lung hypoplasia, facial abnormalities and limb contractures have also been described.
In humans, fetal renal perfusion, which is dependent upon the development of the renin- angiotensin-aldosterone system, begins in the second trimester. Thus, risk to the fetus increases if BLOPRESS PLUS is administered during the second or third trimesters of pregnancy. Animal studies with candesartan cilexetil have demonstrated late fetal and neonatal injury in the kidney. The mechanism is believed to be pharmacologically mediated through effects on the renin- angiotensin-aldosterone system.
Hydrochlorothiazide can reduce the plasma volume as well as the uteroplacental blood flow. It may also cause neonatal thrombocytopenia.

Use in lactation
It is not known whether candesartan cilexetil is excreted in human milk. However, candesartan cilexetil is excreted in the milk of lactating rats. Hydrochlorothiazide passed into mother's milk. Because of the potential for adverse effects on the nursing infant, breast feeding must be discontinued if the use of BLOPRESS PLUS is considered essential (see CONTRAINDICATIONS).

ADVERSE EFFECTS
Clinical Studies
In controlled clinical studies with BLOPRESS PLUS adverse events were mild and transient. Withdrawals from treatment due to adverse events were similar with candesartan cilexetil/hydrochlorothiazide (3.3%) and placebo (2.7%).
In a pooled analysis of clinical trial data, the following common (>1/100) adverse reactions with BLOPRESS PLUS were reported based on an incidence of adverse events with BLOPRESS PLUS at least 1% higher than the incidence seen with placebo:

Frequency/ System Organ Class*	Common
Infections and infestations	Respiratory infection
Nervous system disorders	Dizziness/ vertigo, headache[1]

[1]The ADRs 'respiratory infection' and 'headache' were seen in clinical trials for candersartan cilexetil but were not seen in the clinical trials for BLOPRESS PLUS

Laboratory findings
Increases in serum uric acid, blood glucose and serum ALAT (SGPT) were reported as adverse events slightly more often with BLOPRESS PLUS (crude rates 1.1%, 1.0% and 0.9%, respectively) than with placebo (0.4%, 0.2% and 0% respectively). Minor decreases in hemoglobin and increases in serum ASAT (SGOT) have been observed in patients receiving BLOPRESS PLUS. Increases in creatinine, urea or potassium and decreases in sodium have been observed.
The "very rare" side effects of candesartan cilexetil may include intestinal angioedema of gastrointestinal disorders.

Postmarketing
Following is a list of ADRs which have been observed in postmarketing and are not included above:

Postmarketing for candesartan cilexetil and hydrochlorothiazide
Blood and lymphatic system disorders: Leukopenia, neutropenia and agranulocytosis.
Metabolism and nutrition disorders: Hyponatraemia
Skin and subcutaneous tissue disorders: Rash, urticarial
Renal and urinary disorders: Renal impairment, including renal failure in susceptible patients.

Post –marketing of Candesartan cilexetil
Metabolism and nutrition disorders: Hyperkalemia
Ear and labyrinth disorders: Tinnitus
Respiratory, thoracic and mediastinal disorders: Cough
Hepato-biliary disorders: Increased liver enzymes, abnormal hepatic function or hepatitis.
Skin and subcutaneous tissue disorders: Angioedema, pruritus.
Musculoskeletal, connective tissue and bone disorders: Back pain

Post-marketing of Hydrochlorothiazide
Blood and lymphatic system disorders: Thrombocytopenia, aplastic anemia, bone marrow depression, hemolytic anemia
Immune system disorders: Anaphylactic reactions
Metabolism and nutrition disorders: Hyperglycemia, hyperuricemia, electrolyte imbalance (including hypokalemia)
Psychiatric disorders: Sleep disturbances, depression, restlessness
Nervous system disorders: Light-headedness, paraesthesia
Eye disorders: Transient blurred vision, acute myopia, acute angle-closure glaucoma
Cardiac disorders: Cardiac arrhythmias
Vascular disorders: Postural hypotension, necrotising angitis (vasculitis, cutaneous vasculitis)
Respiratory, thoracic and mediastinal disorders: Respiratory distress (including pneumonitis and pulmonary edema)
Gastrointestinal disorders: Anorexia, loss of appetite, gastric irritation, diarrhea, constipation, pancreatitis
Hepatobiliary disorders: Jaundice (intrahepatic cholestatic jaundice)
Skin and subcutaneous tissue disorders: Photosensitivity reactions, toxic epidermal necrolysis, systemic lupus erythematosus, reactivation of systemic lupus erythematosus, cutaneous lupus erythematosus-like reactions, reactivation of cutaneous lupus erythematosus
Musculoskeletal and connective tissue disorders: Muscle spasm
Renal and urinary disorders: Glycosuria, interstitial nephritis
General disorders and administration site conditions: Weakness, Fever
Investigations: Increases in cholesterol and triglycerides, Increases in BUN and serum creatinine

Neoplasms benign, malignant and unspecified (incl cysts and polyps)
Frequency 'not known': Non-melanoma skin cancer (Basal cell carcinoma and Squamous cell carcinoma)
Description of selected adverse reactions
Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between HCTZ and NMSC has been observed.

PHARMACOLOGICAL PROPERTIES
Pharmacodynamic properties
Candesartan cilexetil/ hydrochlorothiazide:
Pharmaco-therapeutic group: Angiotensin II antagonists + diuretics, ATC code CO9DA06. Candesartan cilexetil

Angiotensin II is the primary vasoactive hormone of the renin-angiotensin-aldosterone system and plays a significant role in the pathophysiology of hypertension and other cardiovascular disorders. It also has an important role in the pathogenesis of organ hypertrophy and end organ damage. The major physiological effects of angiotensin II, such as vasoconstriction, aldosterone stimulation, regulation of salt and water homeostasis and stimulation of cell growth, are mediated via the type 1 (AT1) receptor.

Candesartan cilexetil:
Candesartan cilexetil is a prodrug which is rapidly converted to the active drug, candesartan, by ester hydrolysis during absorption from the gastrointestinal tract. Candesartan is an angiotensin II receptor antagonist, selective for AT1 receptors, with tight binding to and slow dissociation from the receptor. It has no agonist activity.
Candesartan does not influence ACE or other enzyme systems usually associated with the use of ACE inhibitors. Since there is no effect on the degradation of kinins, or on the metabolism of other substances, such as substance P, angiotensin II receptor antagonists are unlikely to be associated with cough. In controlled clinical trials comparing candesartan cilexetil with ACE inhibitors, the incidence of cough was lower in patients receiving candesartan cilexetil. Candesartan does not bind to or block other hormone receptors or ion channels known to be important in cardiovascular regulation. The antagonism of the AT1 receptors results in dose related increases in plasma renin levels, angiotensin I and angiotensin II levels, and a decrease in plasma aldosterone concentration.

The effects of candesartan cilexetil 8-16 mg (mean dose 12 mg) once daily on cardiovascular morbidity and mortality were evaluated in a randomized clinical trial 4,937 elderly patients (aged 70-89 years, 21% aged 80 or above) with mild to moderate hypertension followed for a mean of 3.7 years (Study on Cognition and Prognosis in the Elderly). Patients received candesartan or placebo with other antihypertensive treatment added as needed. The blood pressure was reduced from 166/90 to 145/80 mmHg in the candesartan group, and from 167/90 to 149/82 mmHg in the control group. There was no statistically significant difference in the primary endpoint, major cardiovascular events (cardiovascular mortality, non-fatal stroke and non-fatal myocardial infarction). There were 26.7 events per 1000 patient-years in the candesartan group versus 30.0 events per 1000 patient-years in the control group (relative risk 0.89, 95% CI 0.75 to 1.06, p=0.19).

Hydrochlorothiazide:
Hydrochlorothiazide inhibits the active reabsorption of sodium, mainly in the distal kidney tubules, and promotes the excretion of sodium, chloride and water. The renal excretion of potassium and magnesium increases dose-dependently, while calcium is reabsorbed to a greater extent. Hydrochlorothiazide decreases plasma volume and extracellular fluid and reduces cardiac output and blood pressure. During long-term therapy, reduced peripheral resistance contributes to the blood pressure reduction.

Large clinical studies have shown that long-term treatment with hydrochlorothiazide reduces the risk for cardiovascular morbidity and mortality.

Combination:
Candesartan and hydrochlorothiazide have additive antihyper-tensive effects.
In hypertensive patients, BLOPRESS PLUS causes an effective and long-lasting reduction in arterial blood pressure without reflex increase in heart rate. There is no indication of serious or exaggerated first dose hypotension or rebound effect after cessation of treatment. After administration of a single dose of BLOPRESS PLUS, onset of the antihypertensive effect generally occurs within 2 hours. With continuous treatment, most of the reduction in blood pressure is attained within four weeks and is sustained during long-term treatment. BLOPRESS PLUS once daily provides effective and smooth blood pressure reduction over 24 hours, with little difference between maximum and trough effects during the dosing interval.
In double-blind, randomized studies, the incidence of adverse events, especially cough, was lower during treatment with BLOPRESS PLUS than during treatment with combinations of ACE inhibitors and hydrochlorothiazide.

BLOPRESS PLUS is similarly effective in patients irrespective of age and gender.

Currently there are no data on the use of BLOPRESS PLUS in patients with renal disease/nephropathy, reduced left ventricular function/congestive heart failure and post myocardial infarction.

Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between HCTZ and NMSC has been observed. One study included a population comprised of 71,533 cases of BCC and of 8,629 cases of SCC matched to 1,430,833 and 172,462 population controls, respectively. High HCTZ use (>50,000 mg cumulative) was associated with an adjusted OR of 1.29 (95% CI: 1.23-1.35) for BCC and 3.98 (95% CI: 3.684.31) for SCC. A clear cumulative dose response relationship was observed for both BCC and SCC. Another study showed a possible association between lip cancer (SCC) and exposure to HCTZ: 633 cases of lip-cancer were matched with 63,067 population controls, using a risk-set sampling strategy. A cumulative dose-response relationship was demonstrated with an adjusted OR 2.1 (95% CI: 1.7-2.6) increasing to OR 3.9 (3.0-4.9) for high use (~25,000 mg) and OR 7.7 (5.7-10.5) for the highest cumulative dose (~100, 000 mg).

PHARMACOKINETIC PROPERTIES
Absorption
Candesartan cilexetil:
Following oral administration, candesartan cilexetil is converted to the active drug candesartan. The absolute bioavailability of candesartan is approximately 40% after an oral solution of candesartan cilexetil. The relative bioavailability of a tablet formulation of candesartan cilexetil compared with the same oral solution is approximately 34% with very little variability. The mean peak serum concentration (Cmax) is reached 3-4 hours following tablet intake. The candesartan serum concentrations increase linearly with increasing doses in the therapeutic dose range. There is no accumulation following multiple doses. The area under the serum concentration versus time curve (AUC) of candesartan is not significantly affected by food.

Hydrochlorothiazide:
Hydrochlorothiazide is rapidly absorbed from the gastrointestinal tract with an absolute bioavailability of approximately 70%. Concomitant intake of food increases the absorption by approximately 15%. The bioavailability may decrease in patients with cardiac failure and pronounced edema.

Combination:
Concomitant administration of hydrochlorothiazide and candesartan cilexetil has no clinically significant effect on the pharmacokinetics of each compound. There is an increase in AUC (15-18%) and Cmax (23-24%) of candesartan when given together with hydrochlorothiazide. This is of no clinical importance. No additional accumulation of hydrochlorothiazide or candesartan occurs after repeated doses of the combination compared to individual monotherapies.

Distribution
Candesartan cilexetil:
Candesartan is highly bound to plasma protein (more than 99%). The apparent volume of distribution of candesartan is 0.1 L/kg.

Hydrochlorothiazide:
The plasma protein binding of hydrochlorothiazide is approximately 60%. The apparent volume of distribution is approximately 0.8 L/kg.

Metabolism
Candesartan cilexetil:
Candesartan is mainly eliminated unchanged via urine and bile and only to a minor extent eliminated by hepatic metabolism (CYP2C9). Available interaction studies indicate no effect on CYP2C9 and CYP3A4. Based on in vitro data, no interaction would be expected to occur in vivo with drug whose metabolism is dependent upon cytochrome P450 isozymes CYP1A2, CYP2A6, CYP2C9, CYP2C19, CYP2D6, CYP2E1 or CYP3A4. The half-life of candesartan remains unchanged (approximately 9h) after administration of candesartan cilexetil in combination with hydrochlorothiazide. There is an increase in AUC (15-18%) and Cmax (23-24%) of candesartan when given together with hydrochlorothiazide. This is of no clinical importance. Furthermore titration of the individual components is recommended before switching to BLOPRESS PLUS. No additional accumulation of candesartan occurs after repeated doses of the combination compared to monotherapy.

Hydrochlorothiazide:
Hydrochlorothiazide is not metabolized and is excreted almost entirely as unchanged drug by glomerular filtration and active tubular secretion.

Excretion and Elimination
Candesartan cilexetil:
The terminal half-life ($t_{1/2}$) of candesartan is approximately 9 hours. Total plasma clearance of candesartan is about 0.37 mLl/min/kg, with a renal clearance of about 0.19 mLl/min/kg. The renal elimination of candesartan is both by glomerular filtration and active tubular secretion.
Following an oral dose of 14C-labelled candesartan cilexetil, approximately 26% of the dose is excreted in the urine as candesartan and 7% as an inactive metabolite while approximately 56% of the dose is recovered in the faeces as candesartan and 10% as the inactive metabolite.

Hydrochlorothiazide:
The terminal $t_{1/2}$ of hydrochlorothiazide is approximately 8 hours. Approximately 70% of an oral dose is eliminated in the urine within 48 hours.

Combination:
The $t_{1/2}$ half-life of hydrochlorothiazide remains unchanged (approximately 8 h) after administration of hydrochlorothiazide in combination with candesartan cilexetil. The $t_{1/2}$ of candesartan remains unchanged (approximately 9 h) after administration of candesartan cilexetil in combination with hydrochlorothiazide.

PHARMACOKINETICS IN SPECIAL POPULATIONS
1) Elderly Candesartan cilexetil:
In elderly subjects (over 65 years), Cmax and AUC of candesartan are increased by approximately 50% and 80%, respectively in comparison to young subjects. However, the blood pressure response and the incidence of adverse events are similar after a given dose of BLOPRESS PLUS in young and elderly patients (see DOSAGE AND ADMINISTRATION).
2) Impaired renal function Candesartan cilexetil:
In patients with mild to moderate renal impairment, Cmax and AUC of candesartan increased during repeated dosing by approximately 50% and 70%, respectively, but the terminal $t_{1/2}$ was not altered, compared to patients with normal renal function. The corresponding changes in patients with severe renal impairment were approximately 50% and 110%, respectively. The terminal $t_{1/2}$ of candesartan was approximately doubled in patients with severe renal impairment. The pharmacokinetics in patients undergoing haemodialysis were similar to those in patients with severe renal impairment.

Hydrochlorothiazide:
The terminal $t_{1/2}$ of hydrochlorothiazide is prolonged in patients with renal impairment.

3) Impaired hepatic function Candesartan cilexetil:
In two studies, both including patients with mild to moderate hepatic impairment, there was an increase in the mean AUC of candesartan of approximately 20% in one study and 80% in the other study. There is no experience in patients with severe hepatic impairment

STORAGE
Store below 30°C

SHELF LIFE
3 years

PACKAGING
Box of 28 Tablets

Manufactured by Delpharm Novara S.r.l., Italy

Product Registration Holder:
DKSH MALAYSIA SDN. BHD.
B-11-01, The Ascent, Paradigm,
No.1, Jalan SS 7/26A, Kelana Jaya,
47301 Petaling Jaya, Selangor, Malaysia

Last revision Date:
Nov. 2025 (CCDS v16.0+ Directive)

6201312-01

Mechanical Artwork Identification Panel

General Information

Market:	*MY*
Component Type:	*PI*
Product Name:	*Blopress Plus 16mg/12.5mg*
Material Number:	*6201312-01*
Printing Version No.:	*R01*
Smallest Body Text Size:	*8PT*
Dimension:	*175x600mm*

Version Control

5
Version No.

Mechanical AW History

Date:	Version record and revision description:
02.01.25	V1 - Artwork Build
22.01.25	V2 - Artwork Change
08.05.25	V3 - Artwork Change
27.11.25	V4 - Artwork Change
11.12.25	V5 - Artwork Change
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Colours

- 1. K
- 2. Color Name
- 3. Color Name
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- 9. Color Name
- 10. Color Name

File Name: *PI_6201312-01R01_Blopress Plus_16mg12.5mg_28Tablets_MY_V5*
Account Manager: *Nic. Jiang*
Job Coordinator: *Paul. Wang*
Operator: *Zechen. Yang*